Masterbatches

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
Black Masterbatch (PE Film)	IS 2508 (Polyethylene Materials for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (PE Film); IS 14636 (Flexible Packaging for Food Products); IS 15174 (Carbon Black Masterbatches in Polyethylene resins);IS 16429; IS 11298; IS 11504; IS 12229; IS 13571	ISO 11469 (Plastics — Generic Identification and Marking of Plastics Products); ISO 527-3 (Plastics — Film and Sheeting — Determination of Tensile Properties); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers); ISO 18553 (Assessment of degree of pigment or carbon black dispersion in polyolefin profiles and compounds)	ASTM D1603 (Carbon Black Content in Olefin Plastics); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials, where PE application applies); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D882 (Tensile Properties of Thin Plastic Sheeting); ASTM D1603S (Standard Test Method for Carbon Black Content in Olefin Plastics); ASTM D4218S (Standard Test Method for Determination of Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique); ASTM D6260S (Standard Test Method for Gravimetric Determination of Carbon Black in Nylon Materials (PA)); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 1133-1 (Melt Mass Flow Rate of Thermoplastics); DIN EN ISO 3451-1 (Ash Content of Plastics); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 11664-4 (Colorimetry — CIE Lab* Colour Space)	FDA 21 CFR 178 (Food Contact Additives, where applicable); EU 10/2011; Flexible Packaging Association Guidelines
Black Masterbatch (Pipe Grade)	IS 4984 (HDPE Pipes for Water Supply); IS 14333 (HDPE Pipes for Sewerage and Industrial Applications); IS 15174 (Carbon Black Masterbatches in Polyethylene resins);IS 16429; IS 11298; IS 11504; IS 12229; IS 13571	ISO 4427-1 (Plastics Piping Systems for Water Supply and Drainage — Polyethylene (PE) — General); ISO 1872-1 (Polyethylene Materials); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers); ISO 18553 (Assessment of degree of pigment or carbon black dispersion in polyolefin profiles and compounds)	ASTM D1603 (Carbon Black Content in Olefin Plastics); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D4218 (Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique); ASTM D6260 (Gravimetric Determination of Carbon Black in Nylon Materials)	DIN EN ISO 1872-1 (Polyethylene Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 4892-3 (Artificial Weathering); DIN EN ISO 527-1 (Tensile Properties)	NSF/ANSI 61 (Potable Water Applications, where applicable); PPI Technical Guidelines; Outdoor Polymer Durability Requirements
Black Masterbatch (Cable Grade)	IS 5831 (PVC Insulation and Sheathing Compounds for Electric Cables); IS 7098 (XLPE Power Cables); IS 15174 (Carbon Black Masterbatches in Polyethylene resins); IS 11298 (Polyolefin-based Masterbatches for electrical wire/cable coatings);IS 16429; IS 11504; IS 12229; IS 13571	ISO 1872-1 (Polyethylene Materials); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers); ISO 18553 (Assessment of degree of pigment or carbon black dispersion in polyolefin profiles and compounds)	ASTM D1603 (Carbon Black Content in Olefin Plastics); ASTM D1248 (Polyethylene Plastics Extrusion Materials for Wire and Cable); ASTM D257 (Electrical Resistance of Insulating Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1603S (Standard Test Method for Carbon Black Content in Olefin Plastics); ASTM D4218S (Standard Test Method for Determination of Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique); ASTM D6260S (Standard Test Method for Gravimetric Determination of Carbon Black in Nylon Materials (PA)); ASTM D1238S (Standard Test Method for Melt Flow	DIN EN IEC 60811-501 (Mechanical Tests for Cable Insulation and Sheathing Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 3451-1 (Ash Content); DIN EN ISO 527-1 (Tensile Properties)	UL 94; UL 1581; IEC Electrical Cable Requirements

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
			Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))		
White Masterbatch (TiO_2); Colour Masterbatch (Packaging)	IS 14636 (Flexible Packaging for Food Products); IS 15609 (Multilayer Plastic Pouches for Packaged Drinking Water); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 11469 (Plastics — Generic Identification and Marking of Plastics Products); ISO 22000 (Food safety management systems requirements for raw plastic compounding additives); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D1003 (Haze and Luminous Transmittance of Transparent Plastics); ASTM D2244 (Color Differences of Opaque Materials); ASTM D6290 (Color Determination of Colorant Strength and Tinting Strength); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 11664-4 (Colorimetry); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 3451-1 (Ash Content of Plastics)	FDA 21 CFR 178; EU 10/2011; APR Recycling Design Guidelines (Packaging Applications)
Colour Masterbatch (Injection Moulding)	IS 10910 (Polypropylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 7328 (HDPE for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 11469 (Plastics — Generic Identification and Marking of Plastics Products); ISO 1872-1 (Polyethylene (PE) Materials, where applicable); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D2244 (Color Differences of Opaque Materials); ASTM D6290 (Color Determination of Colorant Strength and Tinting Strength); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 11664-4 (Colour Measurement); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	FDA 21 CFR 178 (Food Contact Grades, where applicable); UL 94 (Electrical Applications); Plastics Industry Colour Standards
Colour Masterbatch (Blow Moulding)	IS 7328 (HDPE for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 12252 (PET for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 11469 (Plastics — Generic Identification and Marking of Plastics Products); ISO 1872-1 (Polyethylene (PE) Materials, where applicable); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D2244 (Color Differences of Opaque Materials); ASTM D6290 (Colorant Strength and Tinting Strength); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials, where PE-based); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 11664-4 (Colour Measurement); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 4892-3 (Weathering Resistance, where outdoor use applies)	FDA 21 CFR 178; EU 10/2011; Packaging Industry Material Requirements
Food Contact Masterbatch	IS 16738 (Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for Safe Use in Contact with Foodstuffs and Pharmaceuticals); IS 9845 (Overall Migration of Plastics in Contact with	ISO 22000 (Food safety management systems requirements for raw plastic compounding additives); ISO 11469 (Plastics — Generic Identification and Marking of Plastics Products); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers); ISO 22002-4 (Prerequisite	ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D792 (Density and Specific Gravity of Plastics); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D6400 (Compostable Plastics, where applicable); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN 1186 (Materials and Articles in Contact with Foodstuffs); DIN EN ISO 10993-5 (Biological Evaluation — Cytotoxicity, where applicable for special applications); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties)	FDA 21 CFR 178; EU 10/2011; NSF/ANSI 51

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Foodstuffs); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	programs on food safety for food packaging and color batch manufacturing)			
Medical Grade Masterbatch	IS 17387 (Packaging Materials for Terminally Sterilized Medical Devices); IS 9845 (Medical/Food Contact Applications); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 10993-1 (Biological Evaluation of Medical Devices — Evaluation and Testing); ISO 13485 (Medical Devices — Quality Management Systems); ISO 10993 (Biological evaluation parameters for medical and pharmaceutical grade polymer colorants); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D638 (Tensile Properties of Plastics); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D792 (Density and Specific Gravity of Plastics); ASTM F2026 (Medical Polymer Applications, where applicable); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 10993-5 (Biological Evaluation of Medical Devices — Cytotoxicity); DIN EN ISO 10993-10 (Tests for Irritation and Skin Sensitization); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties)	USP Class VI; FDA 21 CFR 177; Medical Device Material Requirements
UV Stabilizer Masterbatch; HALS Masterbatch; UV Absorber Masterbatch; Optical Brightener Masterbatch; Blowing Agent Masterbatch	IS 14636 (Flexible Packaging for Food Products) (Packaging Applications); IS 4984 (HDPE Pipe Applications); IS 11504 (Additive Masterbatches (antistatic/slip agent indicators) in Polyolefins);IS 16429; IS 15174; IS 11298; IS 12229; IS 13571	ISO 4892-1 (Plastics — Methods of Exposure to Laboratory Light Sources — General Guidance); ISO 4892-3 (Fluorescent UV Lamps Exposure); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D4329 (Fluorescent UV Exposure of Plastics); ASTM G154 (Operating Fluorescent Ultraviolet Lamp Apparatus for UV Exposure of Nonmetallic Materials); ASTM D2565 (Xenon-Arc Exposure of Plastics Intended for Outdoor Applications); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 4892-2 (Artificial Weathering — Xenon Arc Lamps); DIN EN ISO 4892-3 (Artificial Weathering — Fluorescent UV Lamps); DIN EN ISO 1133-1 (Melt Mass Flow Rate of Thermoplastics); DIN EN ISO 527-1 (Tensile Properties of Plastics)	UL 746C (Outdoor Polymeric Applications); SAE Automotive Weathering Requirements; Agricultural Film Industry Guidelines
Antioxidant Masterbatch	IS 14636 (Flexible Packaging for Food Products); IS 4984 (HDPE Pipes for Water Supply); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 11357-6 (Plastics — Differential Scanning Calorimetry — Oxidation Induction Time); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D3895 (Oxidative-Induction Time of Polyolefins by DSC); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D3350 (Polyethylene Materials, where applicable); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 11357-6 (Oxidation Induction Time of Polymers); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 4892-3 (Weathering Resistance, where applicable)	FDA 21 CFR 178 (Food Contact Additives, where applicable); EU 10/2011 (Food Contact Plastics); Polymer Additives Industry Guidelines
Antiblock Masterbatch; Slip Additive Masterbatch; Antifog Masterbatch; Processing Aid Masterbatch	IS 14636 (Flexible Packaging for Food Products); IS 15609 (Multilayer Plastic Pouches for Packaged Drinking Water); IS 11504 (Additive Masterbatches (antistatic/slip agent indicators) in Polyolefins); IS 13571 (Flame Retardant Additive	Slip Additive Masterbatch; Antifog Masterbatch; Processing Aid Masterbatch;ISO 527-3 (Plastics — Film and Sheeting — Determination of Tensile Properties); ISO 4593 (Plastics — Film and Sheeting — Determination of Thickness); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D1894 (Coefficient of Friction of Plastic Film and Sheeting); ASTM D3359 (Adhesion by Tape Test, where coating/additive performance applies); ASTM D882 (Tensile Properties of Thin Plastic Sheeting); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 8295 (Coefficient of Friction of Plastic Films); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 1183-1 (Density of Plastics)	FDA 21 CFR 178; EU 10/2011; Flexible Packaging Association Guidelines; Packaging Industry Additive Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Masterbatches for engineering polymers);IS 16429; IS 15174; IS 11298; IS 12229				
Antistatic Masterbatch	IS 11504 (Additive Masterbatches (antistatic/slip agent indicators) in Polyolefins);IS 16429; IS 15174; IS 11298; IS 12229; IS 13571	ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D257 (DC Resistance or Conductance of Insulating Materials); ASTM E1549 (Specification for ESD Control Products Used in Cleanrooms, where applicable); ASTM D618 (Conditioning Plastics for Testing); ASTM D638 (Tensile Properties of Plastics); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN 61340-2-3 (Methods for Determining Resistance and Resistivity of Flat Materials); DIN EN 61340-5-1 (Electrostatic Control Protection); DIN EN ISO 1133-1 (Melt Flow Rate); DIN EN ISO 527 (Tensile Properties)	IEC 61340 Series (Electrostatic Control); ANSI/ESD S20.20 (ESD Control Programme); UL 94 (Electrical Applications)
Conductive Carbon Masterbatch	IS 5831 (PVC Insulation and Sheathing Compounds for Electric Cables); IS 15174 (Carbon Black Masterbatches in Polyethylene resins);IS 16429; IS 11298; IS 11504; IS 12229; IS 13571	ISO 11469 (Plastics — Generic Identification and Marking of Plastics Products); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers); ISO 18553 (Assessment of degree of pigment or carbon black dispersion in polyolefin profiles and compounds)	ASTM D257 (DC Resistance or Conductance of Insulating Materials); ASTM D991 (Volume Resistivity of Electrically Conductive and Antistatic Products); ASTM D638 (Tensile Properties of Plastics); ASTM D3350 (Polyethylene Materials, where PE based); ASTM D1603S (Standard Test Method for Carbon Black Content in Olefin Plastics); ASTM D4218S (Standard Test Method for Determination of Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique); ASTM D6260S (Standard Test Method for Gravimetric Determination of Carbon Black in Nylon Materials (PA)); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN 61340-2-3 (Electrical Resistance and Resistivity Measurement); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 1183-1 (Density of Plastics)	IEC 61340 Series; ANSI/ESD S20.20; Electronics Industry Material Specifications
Laser Marking Masterbatch	IS 14636 (Flexible Packaging for Food Products); IS 4984 (HDPE Pipe Applications); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 11469 (Plastics — Generic Identification and Marking of Plastics Products); ISO 11664-1 (Colorimetry — CIE Standard Colorimetric Observers, where colour identification is applicable); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D2244 (Color Differences of Opaque Materials); ASTM D6290 (Color Determination of Colorant Strength and Tinting Strength); ASTM D638 (Tensile Properties of Plastics); ASTM D1003 (Haze and Luminous Transmittance of Transparent Plastics, where applicable); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 11664-4 (Colorimetry — CIE Lab*); DIN EN ISO 4892-3 (Weathering Resistance); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1133-1 (Melt Mass Flow Rate)	IEC 60825 (Laser Safety Applications, where applicable); Automotive OEM Material Specifications; Electronics Industry Requirements
Nucleating Masterbatch; Talc Masterbatch	IS 10910 (Polypropylene and its Copolymers for	ISO 11357-3 (Plastics — Differential Scanning Calorimetry — Determination of Melting and	ASTM D3418 (Transition Temperatures and Enthalpies of Fusion and Crystallization of Polymers by DSC);	DIN EN ISO 11357-3 (DSC — Melting and Crystallization Temperature); DIN EN ISO 3451-1	FDA 21 CFR 178 (Food Contact Additives, where applicable); EU 10/2011; Polymer Processing

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 14636 (Flexible Packaging for Food Products); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	Crystallization Temperatures and Enthalpies); ISO 527-2 (Plastics — Determination of Tensile Properties — Moulding and Extrusion Plastics); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	(Ash Content); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	Industry Guidelines
Clarifying Masterbatch	IS 10910 (Polypropylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 14782 (Plastics — Determination of Haze for Transparent Materials); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D1003 (Haze and Luminous Transmittance of Transparent Plastics); ASTM D5420 (Impact Resistance of Flat Rigid Plastic Specimens); ASTM D3418 (DSC Thermal Properties of Polymers); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 11357-3 (DSC Analysis of Polymers); DIN EN ISO 13468-1 (Determination of Total Luminous Transmittance of Transparent Materials); DIN EN ISO 4892-3 (Weathering Resistance); DIN EN ISO 527 (Tensile Properties)	FDA 21 CFR 178; EU 10/2011; Packaging Industry Material Requirements
Flame Retardant Masterbatch (Polyolefin); Flame Retardant Masterbatch (Engineering Plastics)	IS 17048 (Flame Retardant Plastic Applications); IS 5831 (PVC Insulation and Sheathing Compounds for Electric Cables); IS 13571 (Flame Retardant Additive Masterbatches for engineering polymers);IS 16429; IS 15174; IS 11298; IS 11504; IS 12229	ISO 4589-1 (Plastics — Determination of Burning Behaviour by Oxygen Index — General Requirements); ISO 4589-2 (Determination of Minimum Oxygen Index by Ambient-temperature Test); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers); ISO 18553 (Assessment of degree of pigment or carbon black dispersion in polyolefin profiles and compounds)	ASTM D635 (Rate of Burning and/or Extent and Time of Burning of Plastics); ASTM D3801 (Vertical Burning Characteristics of Solid Plastics); ASTM D2863 (Minimum Oxygen Concentration to Support Combustion of Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 4589-2 (Oxygen Index of Plastics); DIN EN 60695-11-10 (Fire Hazard Testing — Flame Test Methods); DIN EN IEC 60695-2-12 (Glow Wire Flammability Index); DIN EN ISO 527-1 (Tensile Properties)	UL 94 (Flammability Classification); IEC 60695 (Fire Hazard Testing); NFPA Fire Safety Requirements
Desiccant Masterbatch; Barium Sulphate Masterbatch; Odour Scavenger Masterbatch; Moisture Scavenger Masterbatch	IS 14636 (Flexible Packaging for Food Products); IS 15609 (Multilayer Plastic Pouches for Packaged Drinking Water); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 15270 (Plastics — Guidelines for the Recovery and Recycling of Plastics Waste, where moisture scavenging is related to recycled materials); ISO 62 (Plastics — Determination of Water Absorption); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D570 (Water Absorption of Plastics); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D792 (Density and Specific Gravity of Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D6869S (Standard Test Method for Coulometric and Volumetric Determination of Moisture in Plastics Using the Karl Fischer Reaction); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 3451-1 (Ash Content of Plastics); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 1183-1 (Density)	FDA 21 CFR 178 (Food Contact Applications, where applicable); EU 10/2011; Packaging Industry Requirements
Filler Masterbatch ($CaCO_3$)	IS 14636 (Flexible Packaging for Food Products); IS 4984 (HDPE Pipe Applications); IS	ISO 3451-1 (Plastics — Determination of Ash — General Methods); ISO 527-2 (Plastics — Determination of Tensile Properties); ISO 13944 (Analysis of	ASTM D1603 (Carbon Black Content in Olefin Plastics, where applicable); ASTM D5630 (Ash Content in Plastics); ASTM D792 (Density and Specific Gravity of Plastics); ASTM D638 (Tensile	DIN EN ISO 3451-1 (Ash Content of Plastics); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	FDA 21 CFR 178 (Food Contact Applications, where applicable); EU 10/2011; Plastics Processing Industry Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	Properties of Plastics); ASTM E1131S (Standard Test Method for Compositional Analysis by Thermogravimetry (TGA for determining precise pigment, filler, and carrier polymer ratios)); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))		
Anti-microbial Masterbatch	IS 14636 (Flexible Packaging for Food Products); IS 17387 (Medical Device Packaging Applications); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 22196 (Plastics and Other Non-porous Surfaces — Measurement of Antibacterial Activity on Plastics and Other Non-porous Surfaces); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM E2180 (Determining the Activity of Incorporated Antimicrobial Agents in Polymeric or Hydrophobic Materials); ASTM G21 (Resistance of Synthetic Polymeric Materials to Fungi); ASTM D638 (Tensile Properties of Plastics); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 22196 (Measurement of Antibacterial Activity on Plastics and Other Non-Porous Surfaces); DIN EN ISO 20743 (Antibacterial Activity of Textile Products, where textile applications apply); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1133-1 (Melt Mass Flow Rate)	EPA Antimicrobial Product Regulations; FDA 21 CFR 177 (Food Contact Applications); EU Biocidal Product Regulation (BPR)
Oxygen Scavenger Masterbatch	IS 14636 (Flexible Packaging for Food Products); IS 15609 (Multilayer Plastic Pouches for Packaged Drinking Water); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 15105-1 (Plastics — Film and Sheeting — Determination of Gas Transmission Rate — Differential Pressure Method); ISO 15105-2 (Equal-pressure Method); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D3985 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting); ASTM F1927 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting Using Coulometric Sensor); ASTM D882 (Tensile Properties of Thin Plastic Sheeting); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 15105-1 (Gas Transmission Rate Through Plastic Films); DIN EN ISO 15106-1 (Water Vapour Transmission Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1133-1 (Melt Mass Flow Rate)	FDA 21 CFR 177/178 (Food Packaging Applications); EU 10/2011; Active Packaging Industry Guidelines
Bio-based Masterbatch	IS 17088 (Compostable Plastics); IS/ISO 17088 (Specifications for Compostable Plastics); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 16620-1 (Plastics — Biobased Content — General Principles); ISO 16620-2 (Determination of Biobased Carbon Content); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin masterbatch carriers)	ASTM D6866 (Determining the Bio-based Carbon Content of Solid, Liquid, and Gaseous Samples Using Radiocarbon Analysis); ASTM D6400 (Compostable Plastics, where applicable); ASTM D5338 (Aerobic Biodegradation of Plastics Under Controlled Composting Conditions); ASTM D1238S (Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	DIN EN ISO 16620-1 (Plastics — Biobased Carbon Content); DIN EN ISO 16620-2 (Determination of Biobased Carbon Content); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1133-1 (Melt Mass Flow Rate)	USDA BioPreferred Program; ISCC PLUS Certification; TÜV OK Biobased Certification
Compostable Masterbatch	IS 17088 (Specifications for Compostable Plastics); IS 16429; IS 15174; IS 11298; IS 11504; IS 12229; IS 13571	ISO 17088 (Specifications for Compostable Plastics); ISO 14855-1 (Determination of Aerobic Biodegradation of Plastics under Controlled Composting Conditions); ISO 13944 (Analysis of lubricant mixing behaviors modified for auxiliary polyolefin	ASTM D6400 (Compostable Plastics); ASTM D5338 (Aerobic Biodegradation of Plastics Under Controlled Composting Conditions); ASTM D6868 (Biodegradable Plastics Used as Coatings on Paper and Other Compostable Substrates); ASTM D1238S (Standard Test Method for Melt	DIN EN ISO 17088 (Specifications for Compostable Plastics); DIN EN 13432 (Requirements for Packaging Recoverable Through Composting and Biodegradation); DIN EN ISO 14855-1 (Aerobic Biodegradation Under Controlled Composting Conditions); DIN EN	BPI Compostable Certification; TÜV OK Compost Industrial/Home Compost Certification; European Bioplastics Certification Requirements

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
		masterbatch carriers)	Flow Rates of Thermoplastics by Extrusion Plastometer (Crucial for testing masterbatch carrier resin dilution behavior))	ISO 527 (Tensile Properties)	

Source field: Plastic and Plastic Products